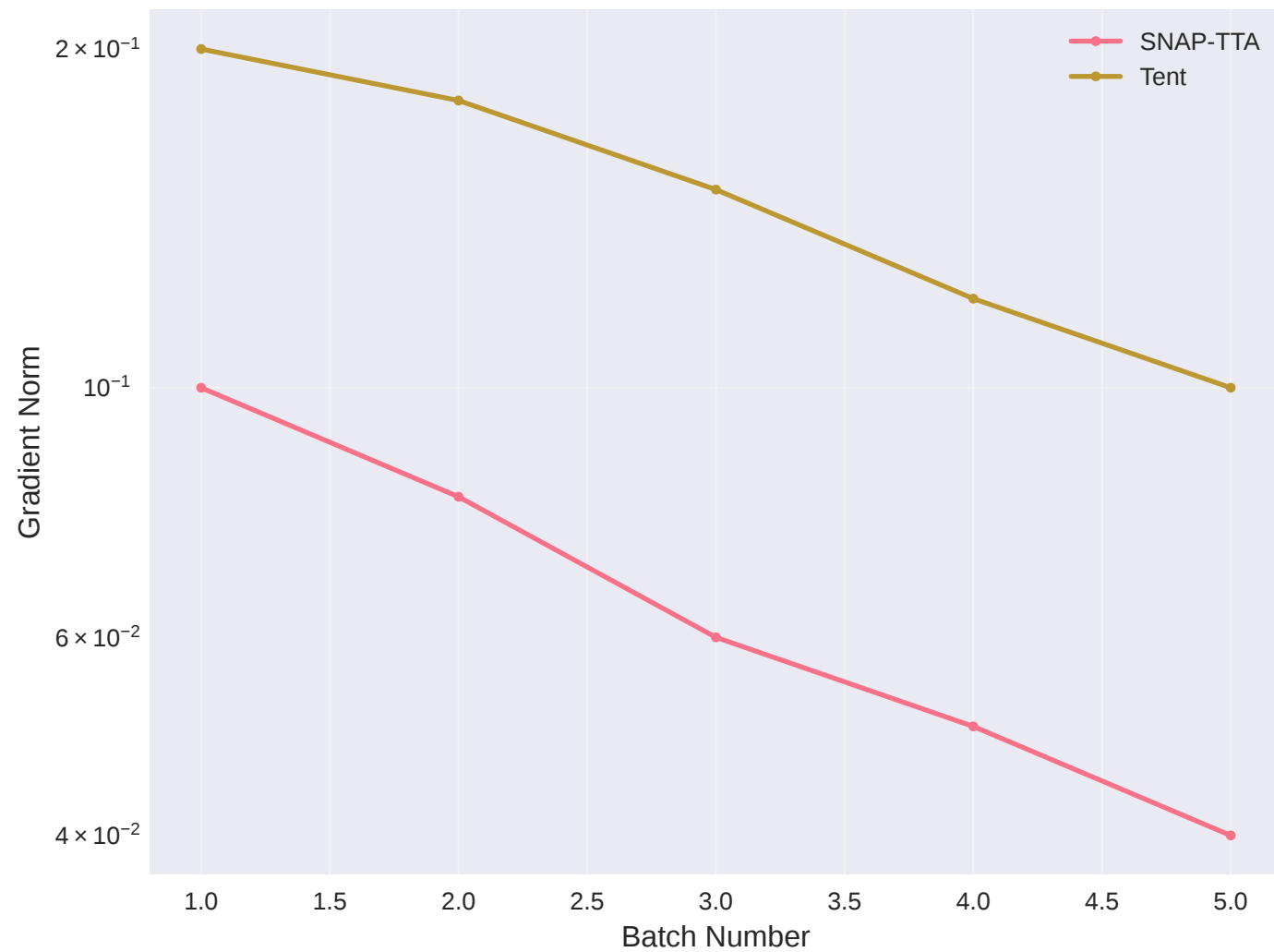
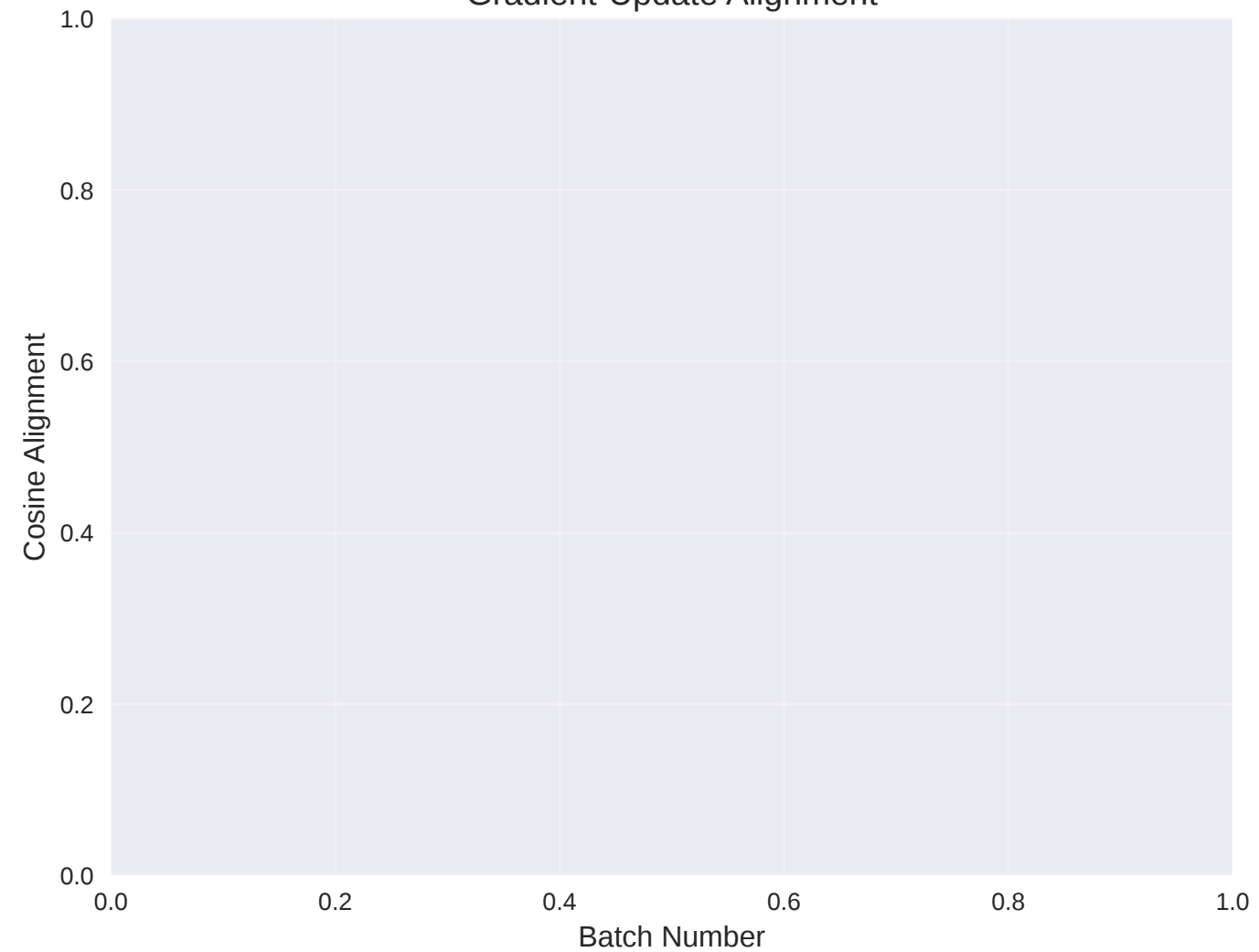


Gradient Analysis

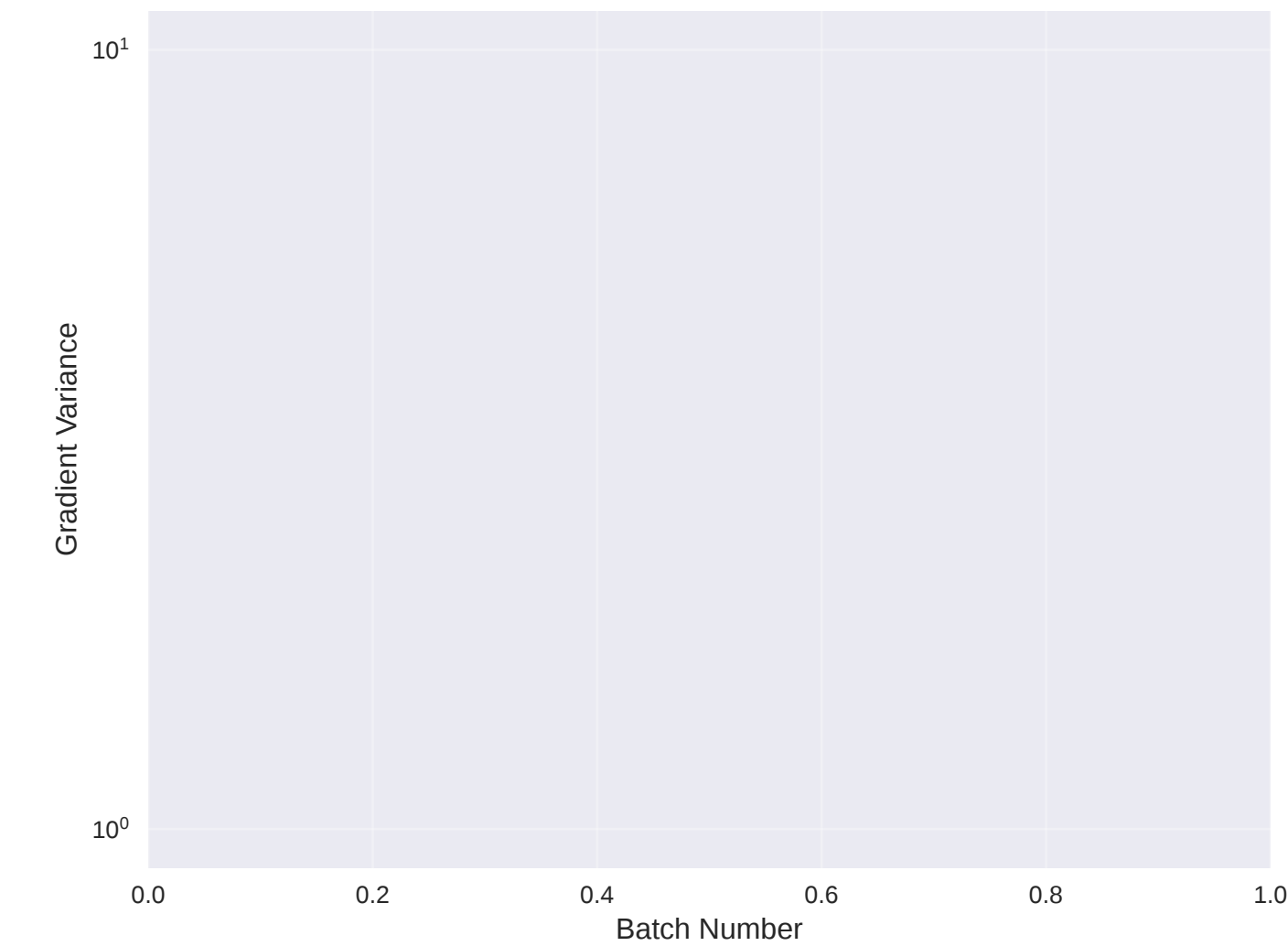
Gradient Norm Evolution



Gradient-Update Alignment



Variance Reduction Effect



Adaptive Learning Rate

